Wavelet Shrinkage Denoising Using the Non-Negative Garrote

Hong-Ye GAO

In this article, we combine Donoho and Johnstone's wavelet shrinkage denoising technique (known as WaveShrink) with Breiman's non-negative garrote. We show that the non-negative garrote shrinkage estimate enjoys the same asymptotic convergence rate as the hard and the soft shrinkage estimates. Simulations are used to demonstrate that garrote shrinkage offers advantages over both hard shrinkage (generally smaller mean-square-error and less sensitivity to small perturbations in the data) and soft shrinkage (generally smaller bias and overall mean-square-error). The minimax thresholds for the non-negative garrote are derived and the threshold selection procedure based on Stein's unbiased risk estimate (SURE) is studied. We also propose a threshold selection procedure based on combining Coifman and Donoho's cycle-spinning and SURE. The procedure is called SPINSURE. We use examples to show that SPINSURE is more stable than SURE: smaller standard deviation and smaller range.

Key Words: Cycle-spinning; Minimax threshold; Nonparametric regression; Shrinkage functions; Stein's unbiased risk estimate (SURE); WaveShrink.

1. INTRODUCTION

Suppose we observe data $\mathbf{y} = (y_1, \dots, y_n)'$ given by

$$y_i = f_i + \sigma z_i \qquad i = 1, 2, \dots, n, \tag{1.1}$$

where $\{f_i\}$ are sampled values of a deterministic function f, which is potentially complex and spatially inhomogeneous, and $\{z_i\}$ are independent Gaussian random variables with mean zero and variance one. Our goal is to estimate f with small L_2 risk (mean-square-error, measured at the observation points)—that is, to find an estimate $\hat{f}$ with small L_2 risk:

$$R(\hat{f},\, f) = \frac{1}{n}\sum_{i=1}^{n} E(\hat{f}_i - f_i)^2. \tag{1.2}$$

Donoho and Johnstone (1994); Donoho and Johnstone (1995); and Donoho, Johnstone, Kerkyacharian, and Picard (1995) developed a methodology called "WaveShrink"

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for estimating f. WaveShrink has very broad asymptotic near-optimality properties. For example, WaveShrink achieves, within a factor of $\log n$, the optimal minimax risk over each functional class in a variety of smoothness classes and with respect to a variety of loss functions, including L_2 risk. WaveShrink is now well established as a technique for *removing noise from corrupted signals*.

WaveShrink is based on the principle of shrinking wavelet coefficients towards zero to remove noise. The WaveShrink estimate is obtained by the following three steps:

1. Transform data $\mathbf{y}$ into the wavelet domain;
2. shrink the empirical wavelet coefficients towards zero; and
3. transform the shrunken coefficients back to the data domain.

Applications of wavelets in statistics can be found, for example, in Antoniadis, Gregoire, and McKeague (1994); Johnstone and Silverman (1997); McCoy and Walden (1996); Nason and Silverman (1994); Ogden (1996); Walter (1994); and Wang (1995).

Shrinkage of the empirical wavelet coefficients works best when the underlying set of the true coefficients of f is *sparse*. That is, when the overwhelming majority of the coefficients of f are small, and the remaining few large ones explain most of the functional form of f. Therefore, an admissible shrinkage function should have the following two ingredients: (1) throw away the small coefficients; and (2) keep the large coefficients.

The shrinkage functions proposed by Donoho and Johnstone are the *hard* and the *soft* shrinkage functions:

$$\delta_\lambda^H(x) = \begin{cases} 0 & |x| \le \lambda \\ x & |x| > \lambda \end{cases} \tag{1.3}$$

$$\delta_\lambda^S(x) = \begin{cases} 0 & |x| \le \lambda \\ x-\lambda & x > \lambda \\ x+\lambda & x < -\lambda, \end{cases} \tag{1.4}$$

where $\lambda \in [0, \infty)$ is the threshold. The top two left-most plots in Figure 1 display the hard and the soft shrinkage functions, respectively.

Asymptotically both hard and soft shrinkage estimates achieve within a factor $\log n$ of the ideal performance (with the aid of an oracle); see Donoho and Johnstone (1994). Finite sample performance was studied by Bruce and Gao (1996b) and Marron et al. (1998). Both the hard and the soft shrinkages have advantages and disadvantages. The soft shrinkage estimates tend to have bigger bias, due to the shrinkage of large coefficients. Due to the discontinuities of the shrinkage function, the hard shrinkage estimates tend to have bigger variance and can be unstable—that is, sensitive to small changes in the data.

To remedy the drawbacks of the hard and the soft shrinkage, we apply non-negative garrote shrinkage to the WaveShrink denoising technique. The non-negative garrote shrinkage function was first introduced by Breiman (1995), and is defined as follows:

$$\delta_\lambda^G(x) = x\left\{1 - (\lambda/x)^2\right\}_+ = \begin{cases} 0 & |x| \le \lambda \\ x - \lambda^2/x & |x| > \lambda. \end{cases} \tag{1.5}$$

A garrote shrinkage function is plotted on the top-right corner of Figure 1. The shrinkage function $\delta_\lambda^G(x)$ is continuous (like the soft shrinkage, therefore more stable than hard)

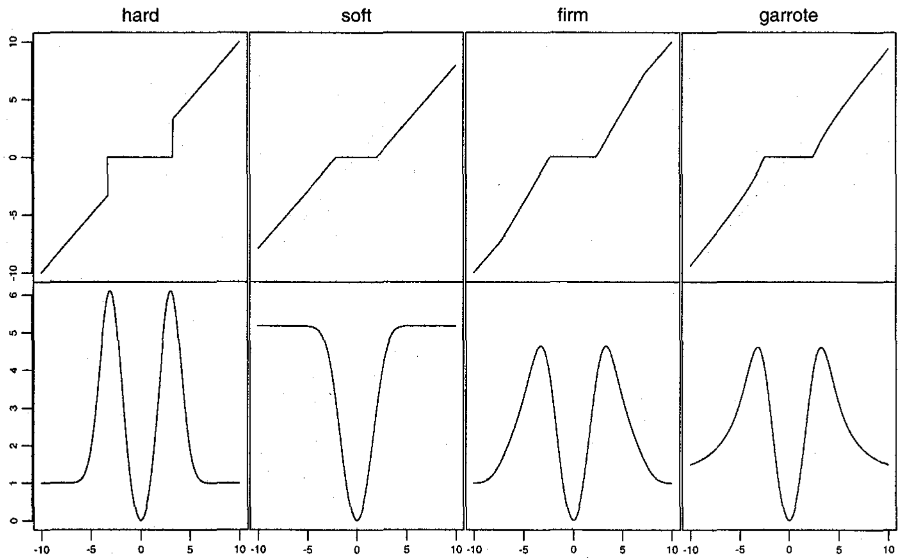

Figure 1. The top row shows shrinkage functions and the bottom row shows the corresponding risk functions defined by (2.1). The vertical dashed lines indicate the thresholds (for hard λ =3.312, for soft λ =2.045, for firm (λ_1, λ_2)=(2.331, 7.259) and for garrote λ =2.441). The hard and the firm shrinkage functions agree with the identity function (dashed lines) for large enough $|x|$. The garrote shrinkage approaches the identity function for large $|x|$. All but the hard shrinkage functions are continuous.

and approaches the identity line as $|x|$ gets large (close to the hard shrinkage, smaller bias than the soft shrinkage for large coefficient). The non-negative garrote shrinkage function provides a good compromise between the hard and the soft shrinkage functions.

Remark 1. *Garrote type of shrinkage was first considered by Strawderman and Cohen (1971) in the context of admissibility of generalized Bayes rules. Breiman (1995) applied this shrinkage technique to subset regression to overcome the drawbacks of stepwise model selection (equivalent to the hard shrinkage in current situations) and ridge regression. In extensive simulation studies, Breiman shows that the non-negative garrote has consistently lower prediction error than subset selection when the true model has many small nonzero coefficients.*

Remark 2. *Gao and Bruce (1997) introduced the firm shrinkage rule $\delta_{\lambda_1,\lambda_2}(\cdot)$:*

$$\delta_{\lambda_1,\lambda_2}(x) = \begin{cases} 0 & \textit{if } |x| \le \lambda_1 \\ sgn(x)\frac{\lambda_2(|x|-\lambda_1)}{\lambda_2-\lambda_1} & \textit{if } \lambda_1 < |x| \le \lambda_2 \\ x & \textit{if } |x| > \lambda_2 \end{cases} . \tag{1.6}$$

Figure 1 displays a firm shrinkage function (top third). By choosing appropriate thresholds (λ_1, λ_2), firm shrinkage outperforms both hard and soft shrinkages. Firm shrinkage has all the benefits of the best of the hard and soft without the drawbacks of either. The only disadvantage of the firm shrinkage is that it requires two thresholds. This makes threshold selection problems much harder and computationally more expensive for adaptive threshold selection procedures such as SURE and cross-validation; see Donoho and Johnstone (1995) and Nason (1995). More details will be given in Section 4.

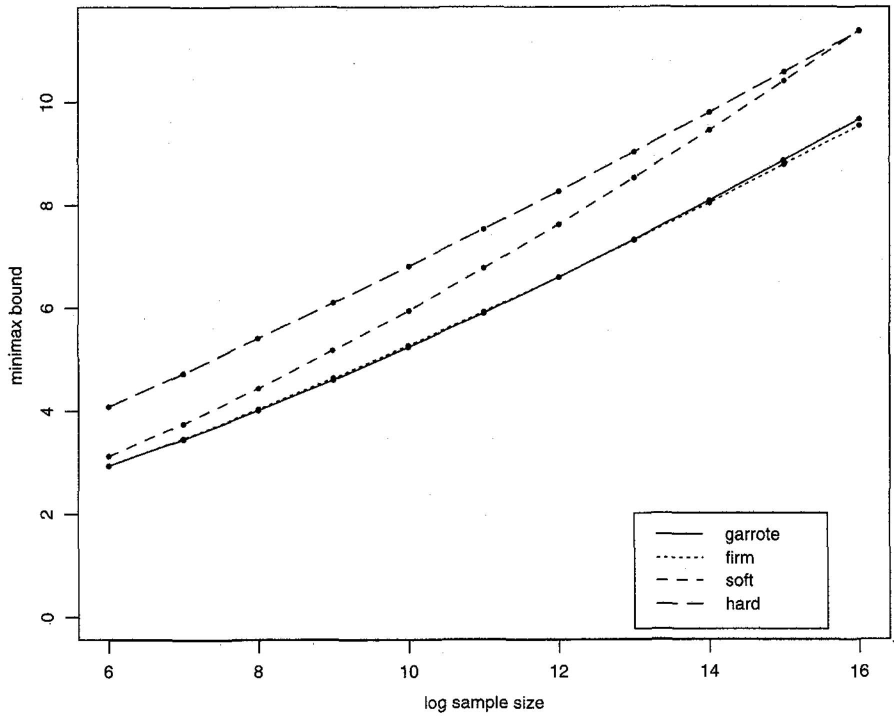

Figure 2. The Minimax Bounds Λ_n^* Defined in (3.3) versus $\log_2(n)$ for Garrote, Firm, Soft, and Hard Shrinkages.

The rest of the article is organized as follows. In Section 2 we give the formulas for computing the bias, the variance, and the L_2 risk (MSE) for the non-negative garrote shrinkage. In Section 3 we compare, theoretically and numerically, the non-negative garrote with the hard, the soft and the firm shrinkage. In Section 4 we discuss the issue of threshold selection. We derive the minimax thresholds for the non-negative garrote, study the SURE (Stein's unbiased risk estimate) threshold selection procedure and a more stable procedure called SPINSURE which combines SURE and "cycle-spinning." Section 5 gives a summary and discussion. Proofs and technical details are given in Section 6.

2. BIAS, VARIANCE, AND L_2 RISK FOR GARROTE SHRINKAGE

In this section, we give formulas for the bias, variance, and L_2 risk of non-negative garrote shrinkage estimate of a Gaussian random variable.

Let $X \sim N(\theta,\ 1)$. Under shrinkage function $\delta_\lambda(\cdot)$ and threshold λ, define the mean, variance and L_2 risk functions of the shrinkage estimator of θ by

$$
\begin{aligned}
M_\lambda(\theta) &= E\{\delta_\lambda(X)\} \\
V_\lambda(\theta) &= \operatorname{var}\{\delta_\lambda(X)\}
\end{aligned}
$$

$$R_\lambda(\theta) = E\{\delta_\lambda(X)-\theta\}^2 = V_\lambda(\theta) + \left(M_\lambda(\theta)-\theta\right)^2. \qquad (2.1)$$

We will use superscripts "F", "G", "H", and "S" to represent the "firm", the "garrote", the "hard", and the "soft", respectively. Formulas for the hard, the soft and the firm shrinkage can be found in Bruce and Gao (1996b) and Gao and Bruce (1997). By straightforward calculus, we have the following:

Proposition 1. *Let Φ and ϕ be the probability distribution and the density functions for standard Gaussian random variable, respectively. Define*

$$A_\lambda(\theta) = \int_\lambda^\infty \left\{\frac{\phi(x-\theta)-\phi(x+\theta)}{x}\right\} dx \qquad (2.2)$$

$$B_\lambda(\theta) = \int_\lambda^\infty \left\{\frac{\phi(x-\theta)+\phi(x+\theta)}{x^2}\right\} dx. \qquad (2.3)$$

Then

$$\begin{aligned} M_\lambda^G(\theta) &= M_\lambda^H(\theta) - \lambda^2 A_\lambda(\theta) \\ V_\lambda^G(\theta) &= V_\lambda^H(\theta) + \lambda^4\left(B_\lambda(\theta) - A_\lambda(\theta)^2\right) + 2\lambda^2 A_\lambda(\theta) M_\lambda^H(\theta) \\ &\quad -2\lambda^2\{1 - \Phi(\lambda-\theta) + \Phi(-\lambda-\theta)\} \\ R_\lambda^G(\theta) &= R_\lambda^H(\theta) + 2\theta\lambda^2 A_\lambda(\theta) + \lambda^4 B_\lambda(\theta) \\ &\quad -2\lambda^2\{1 - \Phi(\lambda-\theta) + \Phi(-\lambda-\theta)\}, \end{aligned} \qquad (2.4)$$

where

$$M_\lambda^H(\theta) = \theta + \theta\{1 - \Phi(\lambda-\theta) - \Phi(\lambda+\theta)\} + \phi(\lambda-\theta) - \phi(\lambda+\theta),$$

$$\begin{aligned} V_\lambda^H(\theta) = (\theta^2+1)\{2 - \Phi(\lambda-\theta) - \Phi(\lambda+\theta)\} + (\lambda+\theta)\phi(\lambda-\theta) \\ +(\lambda-\theta)\phi(\lambda+\theta) - M_\lambda^H(\theta)^2, \end{aligned}$$

and

$$\begin{aligned} R_\lambda^H(\theta) = 1 + (\theta^2-1)\{\Phi(\lambda-\theta) - \Phi(-\lambda-\theta)\} \\ +(\lambda+\theta)\phi(\lambda+\theta) + (\lambda-\theta)\phi(\lambda-\theta) \end{aligned}$$

are the mean, variance, and L_2 risk functions for the hard shrinkage.

Functions $A_\lambda(\theta)$ and $B_\lambda(\theta)$ can only be evaluated numerically. The computational formulas are given in Section 5. The risk functions for the hard and the soft shrinkage can be found in Bruce and Gao (1996b); Donoho and Johnstone (1994); and Marron et al. (1996); and the risk function for firm shrinkage can be found in Gao and Bruce (1997). Risk functions for the hard, the soft, the firm, and the garrote shrinkages are plotted on the bottom row of Figure 1. Like the hard and the firm shrinkage functions, $R_\lambda^G(\theta) \to 1$ as $|\theta| \to \infty$.

Now we turn our attention to the risk calculation of WaveShrink estimates. Let J be the number of resolution levels in the wavelet transform and denote the empirical wavelet coefficients by $\mathbf{w} = (\mathbf{d}_1, \ldots, \mathbf{d}_J, \mathbf{s}_J)$, where $\mathbf{d}_1$ are the finest (resolution) level detail coefficients (a vector of length $n/2$, obtained by filtering $\mathbf{y}$ and dyadic decimation), $\mathbf{d}_J$ are the coarsest level detail coefficients (a vector of length $n/2^J$), and $\mathbf{s}_J$ are the coarsest level smooth coefficients. Details on wavelet transform can be found in many books—Daubechies (1992); Walter (1994); Wickerhauser (1994); Bruce and Gao (1996a); and Ogden (1996), for example. For the global L_2 risk of WaveShrink estimate defined in (1.2), we have

Proposition 2. *Under model (1.1) and an orthogonal wavelet transform,*

$$R(\hat{f}_\lambda, f) = \frac{\sigma^2}{2^J} + \frac{\sigma^2}{n}\sum_{j=1}^{J}\sum_k R_\lambda\left(\frac{\theta_{j,k}}{\sigma}\right), \tag{2.5}$$

where $\Theta = \{\theta_{j,k}\}$ *are the true detail wavelet coefficients of* $\{f_i\}$. The proof is straightforward from (1.2), (2.1), and the orthogonality of the wavelet transform. Formula (2.5) is used for risk calculation throughout the article.

3. COMPARISONS WITH OTHER SHRINKAGE RULES

In this section, we compare the non-negative garrote with the hard, the soft, and the firm shrinkage schemes. We will mainly focus on three areas: (1) the garrote shrinkage achieves the same asymptotic convergence rate as the soft and the hard shrinkage (within a factor $\log n$ of the ideal risk); (2) finite sample performance with "ideal" thresholds—thresholds selected to minimize (2.5) where the underlying function is assumed to be known (this method is called the "crystal-ball method" by Breiman (1995) and the L_2 risks with the "ideal" thresholds will be called "crystal-ball risk"); and (3) finite sample performance with the minimax thresholds. Donoho and Johnstone's four signals: "blocks", "bumps", "heavisine", and "doppler" are used in (2) and (3). The definitions of the four signals can be found in Donoho and Johnstone (1994). The top row of Figure 3 display the noiseless signals (solid lines). All computations were carried out using S+Wavelets version 1.1.

Other comparisons such as local shift sensitivity and bias reduction can be carried out the same way as in Gao and Bruce (1997). It can be shown that the garrote shrinkage, like the firm shrinkage, is less sensitive than the hard shrinkage to small fluctuations in the data; and less biased than the soft shrinkage.

3.1 ASYMPTOTIC NEAR-OPTIMALITY

We show that the Theorem 1 of Donoho and Johnstone (1994) holds for the garrote shrinkage. Let

$$R_{\text{oracle}}(\Theta) = \frac{\sigma^2}{2^J} + \frac{1}{n}\sum_{j=1}^{J}\sum_k \left(\theta_{j,k}^2 \wedge \sigma^2\right) \tag{3.1}$$

be the oracle furnished risk (so called ideal risk which is practically unattainable, details can be found in Donoho and Johnstone 1994, sec. 2.1). This ideal risk is usually of order n^{-r} for some $r < 1$, uniformly over a large collection of functions, including smooth and nonsmooth functions simultaneously; for example, certain Besov space, see Donoho

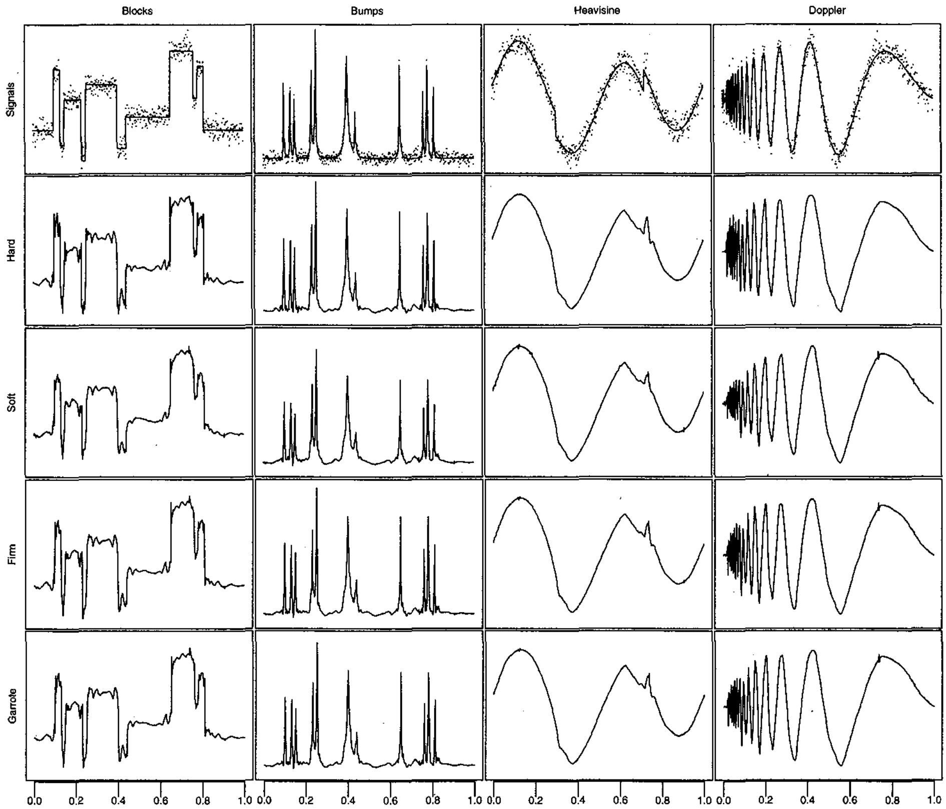

Figure 3. Donoho and Johnstone's Four Examples. Top row: noiseless (lines, n =512) and noisy (dots, SNR=5) signals; second row: WaveShrink estimates with hard shrink and minimax threshold λ =3.312; third row: WaveShrink estimates with soft shrink and minimax threshold λ =2.045; fourth row: WaveShrink estimates with firm shrink and minimax thresholds (λ_1, λ_2) = (2.331, 7.259); bottom row: WaveShrink estimates with non-negative garrote and minimax threshold λ =2.441; s8 wavelet was used for all signals.

and Johnstone (1998) for more details. Donoho and Johnstone have shown that for both the hard and the soft shrinkages, the asymptotic L_2 risk is within a factor of $\log n$ of the ideal risk. The following theorem shows that the same holds for the garrote shrinkage. The proof of the theorem is in the Appendix.

Theorem 1. *Under model (1.1) and the orthogonal wavelet transform,*

$$R(\hat{f}_\lambda^G, f) \leq (\lambda^2 + 3)\left\{R_{\text{oracle}}(\Theta) + \frac{4\sigma^2\phi(\lambda)}{\lambda}\right\}. \tag{3.2}$$

Table 1. Minimax Thresholds for Non-negative Garrote

n	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
λ_n^*	1.883	2.075	2.261	2.441	2.615	2.784	2.948	3.107	3.262	3.413	3.561
Λ_n^*	2.934	3.452	4.008	4.603	5.233	5.899	6.596	7.324	8.079	8.861	9.666

Table 2. Minimax Thresholds Used in Calculations

n	*128*	*256*	*512*	*1024*	*2048*
Hard: λ	2.913	3.117	3.312	3.497	3.674
Soft: λ	1.669	1.859	2.045	2.226	2.403
Firm: λ_1	1.893	2.116	2.331	2.538	2.737
Firm: λ_2	7.980	7.549	7.259	7.069	6.939
Garrote: λ	2.075	2.261	2.441	2.615	2.784

For $\lambda = \sqrt{2\alpha \log n}$,

$$R(\hat{f}^G_\lambda, f) \leq C \log n \left\{ R_{\text{oracle}}(\Theta) + O\left(\frac{1}{n^\alpha \sqrt{\log n}}\right)\right\}.$$

Hence, for appropriately chosen $\alpha \in (0, 1]$, *non-negative garrote shrinkage estimate achieves within a factor log* n *of the ideal risk.*

3.2 Finite Sample Comparison: Crystal-Ball Risk

This section compares the crystal-ball risk using all four shrinkage methods by selecting λ to minimize (2.5) where the underlying function is assumed to be known.

The noisy signals are generated by adding Gaussian white noise. The standard deviation (SD) of the noise is determined by signal-to-noise ratio (SNR). The SNR is defined as

$$\text{SNR} = \frac{\text{SD(signal)}}{\text{SD(noise)}}.$$

Throughout our numerical study, we choose SNR $= 5$. Figure 3 displays the noiseless, noisy, and estimated signals (using minimax thresholds, see next section for more details) for $n = 512$. Daubechies' "least asymmetric" wavelet *S8* (with 8 low-pass filter coefficients) is used. The ideal thresholds are computed from (2.5) using grid search over λ with an increment of $\Delta = .01$ and over all resolution levels.

The obtained crystal-ball risks are tabulated in Table 3 and the corresponding thresholds are listed in Table 4. We have the following observations:

1. Among the 20 combinations of sample size/signal/wavelet, the garrote shrinkage is better (smaller L_2 risks) than the hard shrinkage for all cases.
2. Among the 20 combinations of sample size/signal/wavelet, the garrote shrinkage is better (smaller L_2 risks) than the soft shrinkage all but two cases: for the heavisine signal and $n \leq 256$.
3. Ideal garrote shrinkage is comparable to ideal firm shrinkage, while the firm shrinkage requires two thresholds.

3.3 Finite Sample Comparison: Minimax Risk

We compare the minimax risks among the four shrinkage functions. The minimax threshold λ^* was introduced by Donoho and Johnstone (1994) and is defined as the λ

which minimizes the quantity

$$\Lambda_n^* \equiv \inf_\lambda \sup_\theta \left\{ \frac{R_\lambda(\theta)}{n^{-1} + (\theta^2 \wedge 1)} \right\}. \tag{3.3}$$

Λ_n^* is called the minimax risk bound. The minimax thresholds for the soft shrinkage were derived in Donoho and Johnstone (1994); the minimax thresholds for the hard shrinkage were computed by Bruce and Gao (1996a); and the minimax thresholds for the firm shrinkage can be found in Gao and Bruce (1997).

Table 3. The Crystal-Ball Risks are Computed by Optimizing Over all Resolution Levels. The risks for garrote, hard and soft are obtained by grid search over λ with $\Delta = .01$, and the firm risks are by grid $\Delta = (.01, .01)$. The minimax thresholds are listed in Table 2 and the corresponding risks are computed by shrinking all but 4 coarsest levels (all but 16 coefficients). The noise variance is 1 and the signals are rescaled so that the SNR is 5. *S8* wavelet is used for all examples. For given sample size and signal, the lowest risk is indicated by two asterisks and the second lowest risk is indicated by one asterisk.

Signal		*Shrinkage*	*n = 128*	*n = 256*	*n = 512*	*n = 1024*	*n = 2048*
Blocks	Crystal Ball	Garrote	.702**	.539**	.384**	.268**	.181**
Blocks	Crystal Ball	Firm	.708*	.551*	.395*	.275*	.185*
Blocks	Crystal Ball	Hard	.769	.613	.428	.300	.196
Blocks	Crystal Ball	Soft	.730	.579	.420	.305	.214
Blocks	Minimax	Garrote	.926	.725	.518	.351	.244
Blocks	Minimax	Firm	.956	.724	.505	.335	.231
Blocks	Minimax	Hard	.973	.762	.530	.352	.242
Blocks	Minimax	Soft	1.172	.974	.739	.533	.386
Bumps	Crystal Ball	Garrote	.677*	.579**	.437*	.299**	.194**
Bumps	Crystal Ball	Firm	.666**	.589*	.435**	.302*	.196*
Bumps	Crystal Ball	Hard	.765	.664	.448	.331	.208
Bumps	Crystal Ball	Soft	.727	.629	.519	.375	.255
Bumps	Minimax	Garrote	.793	.775	.580	.397	.256
Bumps	Minimax	Firm	.780	.768	.554	.370	.238
Bumps	Minimax	Hard	.825	.815	.565	.400	.250
Bumps	Minimax	Soft	1.166	1.120	.960	.688	.470
Heavisine	Crystal Ball	Garrote	.263	.174	.117*	.071*	.042*
Heavisine	Crystal Ball	Firm	.256**	.172**	.117**	.071**	.041**
Heavisine	Crystal Ball	Hard	.290	.199	.135	.076	.043
Heavisine	Crystal Ball	Soft	.256*	.172*	.120	.075	.047
Heavisine	Minimax	Garrote	.265	.174	.117	.072	.044
Heavisine	Minimax	Firm	.261	.173	.117	.072	.043
Heavisine	Minimax	Hard	.309	.202	.135	.079	.045
Heavisine	Minimax	Soft	.257	.172	.120	.081	.054
Doppler	Crystal Ball	Garrote	.506*	.354*	.259*	.174*	.107*
Doppler	Crystal Ball	Firm	.498**	.347**	.256**	.172**	.103**
Doppler	Crystal Ball	Hard	.560	.399	.306	.196	.115
Doppler	Crystal Ball	Soft	.565	.413	.307	.216	.140
Doppler	Minimax	Garrote	.548	.385	.291	.196	.119
Doppler	Minimax	Firm	.551	.379	.284	.191	.114
Doppler	Minimax	Hard	.567	.403	.323	.206	.119
Doppler	Minimax	Soft	.729	.530	.398	.281	.185

The minimax thresholds for the non-negative garrote were computed from (3.3) and tabulated in Table 1. The values in Table 1 were computed using a grid search over λ with increments $\Delta_\lambda = .00001$. At each grid point, the supremum over θ was computed using S-Plus nonlinear minimization function *nlmin()*.

The obtained minimax bounds are also listed in Table 1 and plotted along with minimax bounds for the soft, the hard, and the firm in Figure 2. The minimax bounds for the soft, the hard, and the firm are all listed in Gao and Bruce (1997, tab. 1). From Figure 2, we can see that garrote has tighter minimax bounds than both the hard and the soft shrinkage rules, and comparable to the firm shrinkage rule. Numerical comparisons with other shrinkage schemes can be found in Table 3.

The minimax thresholds used in our simulation are listed in Table 2. As in the previous section, Daubechies' "least asymmetric" wavelet *S8* is used. Wavelet coefficients in all but the coarsest four levels are shrunk (all but 16 coefficients).

Results are summarized in Table 3. From the table, we can see that non-negative garrote performs well when sample size is small. Among the 20 signal/wavelet/sample-size combinations, the garrote shrinkage is better than the hard shrinkage in 16 cases and better than the soft shrinkage in 18 cases.

Remark 3. *Donoho and Johnstone have shown that using a "universal" threshold ($\sqrt{2\log(n)}$) will thoroughly suppress noise and achieve asymptotically near-optimal convergence rate, regardless of the underlying signal (as long as it is in a certain Besov space). Minimax threshold, which is derived based on the same principle (optimize a minimax bound, not the minimax risk itself) and is smaller than universal threshold, can be viewed as a fine-tuned universal threshold. In practice, both universal and minimax threshold tend to be bigger than necessary, therefore over smooth the underlying signal. From Table 4, we can see that except for the "heavisine" signal, which is quite smooth, the minimax thresholds are all bigger than the ideal crystal-ball thresholds. For a given signal, especially a nonsmooth signal, minimax threshold only guarantees to remove noise, but it may also remove some features at the same time. A data-driven threshold selection procedure is desirable to preserve more features. This is our topic for the next section.*

Remark 4. *In practice, which wavelet to use and how many levels to compute are very important aspects of WaveShrink. Generally they are determined by the sparsity of the resulting wavelet coefficients. The sparser the wavelet coefficients, the fewer coefficients needed to represent the original signal, and therefore the more efficiently the procedure works. Sparsity can be measured by L_1-norm, or entropy related measures, see Wickerhauser (1994, chap. 8) for more measures.*

Remark 5. *In both Section 3.2 and 3.3, the results are analytic and do not require any simulation.*

4. EMPIRICAL THRESHOLD SELECTION

From previous section, we have seen that minimax thresholds are generally bigger than the ideal crystal-ball thresholds, especially for nonsmooth signals. Data-driven empirical threshold selection procedures are highly desirable. In this article, we study two

Table 4. The Thresholds for Crystal-Ball Method, SURE, and SPINSURE. The mean, minimum and maximum SURE and SPINSURE thresholds are based on 100 simulations. Minimax thresholds and the "universal" thresholds ($\sqrt{2\log(n)}$) are listed at the bottom of the table as references.

Signal		$n = 128$	$n = 256$	$n = 512$	$n = 1024$	$n = 2048$
	Crystal Ball	1.23	1.49	1.74	1.93	2.07
	SPINSURE: mean	1.32	1.51	1.73	1.89	2.08
	SPINSURE: min	.75	1.05	1.37	1.58	1.82
Blocks	SPINSURE: max	1.85	1.82	1.97	2.15	2.32
	SURE: mean	1.14	1.41	1.65	1.79	1.98
	SURE: min	.46	.90	.93	1.26	1.64
	SURE: max	1.84	1.80	2.09	2.26	2.31
	Crystal Ball	1.35	1.43	1.63	1.87	2.07
	SPINSURE: mean	1.23	1.43	1.63	1.84	2.06
	SPINSURE: min	.66	1.06	1.26	1.57	1.82
Bumps	SPINSURE: max	1.67	1.82	1.93	2.12	2.31
	SURE: mean	1.23	1.35	1.56	1.78	1.97
	SURE: min	.44	.85	1.16	1.29	1.58
	SURE: max	1.94	1.82	2.14	2.34	2.38
	Crystal Ball	2.23	2.28	2.34	2.60	2.75
	SPINSURE: mean	2.15	2.28	2.39	2.52	2.68
	SPINSURE: min	1.51	1.68	1.93	2.14	2.30
Heavisine	SPINSURE: max	2.68	2.74	2.86	2.90	3.10
	SURE: mean	1.95	2.13	2.18	2.39	2.51
	SURE: min	1.01	1.33	1.43	1.79	1.76
	SURE: max	2.72	3.10	2.87	2.98	3.03
	Crystal Ball	1.60	1.79	1.90	2.08	2.29
	SPINSURE: mean	1.57	1.76	1.93	2.10	2.33
	SPINSURE: min	1.14	1.37	1.47	1.77	2.07
Doppler	SPINSURE: max	2.00	2.08	2.22	2.47	2.63
	SURE: mean	1.54	1.69	1.89	2.02	2.25
	SURE: min	.88	.83	1.36	1.51	1.76
	SURE: max	2.27	2.33	2.43	2.52	2.75
Minimax		2.08	2.26	2.44	2.62	2.78
Universal		3.12	3.33	3.53	3.72	3.91

threshold selection procedures: the first one is based on Stein's unbiased risk estimate (SURE); and the second one is based on combining cycle-spinning and SURE.

4.1 SURE PROCEDURE

For soft shrinkage, Donoho and Johnstone (1995) proposed the SURE (Stein's unbiased risk estimate) threshold selection procedure; and Nason (1995) studied a cross-validation procedure. These empirical thresholds are data-driven and therefore adapt to particular situations. We will derive the SURE procedure for the garrote shrinkage. Cross-validation procedure can be carried out in a fashion similar to Nason (1995). Note that hard shrinkage function is not continuous, so it does not have bounded weak derivative (in Stein's sense, see Stein 1981, def. 1), and therefore SURE procedure for hard shrinkage is not valid.

Since the non-negative garrote shrinkage function $\delta_\lambda^G(x)$ is weakly differentiable in

Stein's sense (Stein 1981) with

$$\nabla \delta_\lambda^G(x) = \left(1 + \frac{\lambda^2}{x^2}\right) I(|x| > \lambda),$$

where $I(\cdot)$ is the indicator function, the L_2 risk (2.1) can be estimated unbiasedly by

$$S_\lambda^G(X) \equiv 1 + (X^2 - 2) I(|X| \le \lambda) + \frac{\lambda^4 + 2\lambda^2}{X^2} I(|X| > \lambda). \tag{4.1}$$

Indeed, let $X \sim N(\theta,\ 1)$, by straightforward calculus,

$$E\left\{S_\lambda^G(X)\right\} = R_\lambda^G(\theta) = E\left\{\delta_\lambda^G(X) - \theta\right\}^2.$$

For the global L_2 risk defined in (1.2) and Proposition 2, we have the following.

Proposition 3. *Under model (1.1) and orthogonal wavelet transform,*

$$SURE(\boldsymbol{\lambda};\ \mathbf{w}) = \frac{\sigma^2}{2^J} + \frac{\sigma^2}{n} \sum_{j=1}^{J} \sum_k S_{\lambda_j}^G\left(\frac{d_{j,k}}{\sigma}\right) \tag{4.2}$$

is an unbiased estimate of risk $R(\hat{f}_{\boldsymbol{\lambda}},\ f)$, where $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_J)$ are level-dependent thresholds.

The SURE thresholds are obtained by

$$\boldsymbol{\lambda}^* = \arg\min_{\boldsymbol{\lambda}} \left\{\mathrm{SURE}(\boldsymbol{\lambda};\ \mathbf{w})\right\}. \tag{4.3}$$

For the soft shrinkage, a simple case, $\lambda_j = \lambda$ for all $j = 1, \ldots, J$, has been studied by Nason (1995). The threshold derived from (4.3) is called the global SURE threshold. Our numerical studies are all focused on the global SURE threshold. For the global SURE procedure, a fast algorithm can be derived from the following:

Proposition 4. *Let $(x_1, \ldots, x_n)$ be a vector and*

$$R_n(\lambda) = \sum_{i=1}^{n} S_\lambda^G(x_i).$$

Then

$$\arg\min_{\lambda > 0} R_n(\lambda) = \arg\min_{\lambda \in \{|x_i|\}} R_n(\lambda).$$

Proof: Without loss of generality, we assume that $\{x_i\}$ are ordered and non-negative (otherwise replace $\{x_i\}$ by ordered $\{|x_i|\}$). For $x_k \le \lambda < x_{k+1}$, from (4.1),

$$\begin{aligned} R_n(\lambda) &= n - 2\sum_{i=1}^{n} I(|x_i| \le \lambda) + \sum_{i=1}^{n} x_i^2 I(|x_i| \le \lambda) - (\lambda^4 + 2\lambda^2) \sum_{i=1}^{n} x_i^{-2} I(|x_i| > \lambda) \\ &= n - 2k + \sum_{i=1}^{k} x_i^2 + (\lambda^4 + 2\lambda^2) \sum_{i=k+1}^{n} x_i^{-2} \end{aligned}$$

is increasing in λ and therefore

$$\arg \min_{x_k \leq \lambda < x_{k+1}} R_n(\lambda) = x_k.$$

□

We code the fast algorithm in S language:

```
gsure ¡- function(x)
–
   n ¡- length(x)
   sx ¡- sort(x^2)
   cx ¡- cumsum(sx)
   nx ¡- n - 2*(1:n)
   ix ¡- (sx^2 + 2*sx)*rev(c(0, cumsum(1/rev(sx))[-n]))
   rk ¡- (nx + cx + ix)/n
   sqrt(sx[order(rk)[1]])
”
```

Remark 6. *Since in WaveShrink, we do not shrink all the coefficients, we should only apply gsure to the shrunk coefficients.*

4.2 SPINSURE PROCEDURE

Stationary wavelet transform (also known as nondecimated wavelet transform or translation-invariant wavelet transform, even though it is really translation-equivariant) was used by Lang et al. (1995) and Coifman and Donoho (1995) to overcome the drawbacks of orthogonal (maximally-decimated) wavelet transforms and to produce more stable estimates. It provides a natural way to generate multiple estimates for the same object, without doing any re-sampling. Their procedure is based on "Cycle-Spinning":

$$\text{Shift} \rightarrow \text{WaveShrink} \rightarrow \text{Unshift} \rightarrow \text{Combine.}$$

The same idea, combined with SURE, can be used for threshold selection. We shall call the following procedure "SPINSURE".

1. Shift data $\mathbf{y}$ by j: for example,

$$\mathbf{y}_j = \begin{cases} (y_{j+1}, \ldots, y_n, y_1, \ldots, y_j)' & \text{if } j \geq 0, \\ (y_{n+j+1}, \ldots, y_n, y_1, \ldots, y_{n+j})' & \text{if } j < 0; \end{cases}$$

2. Apply SURE to the shifted data to get λ_j;
3. Combine thresholds $\{\lambda_j\}$.

There are many ways to combine $\{\lambda_j\}$. Obvious choices include mean, median, quantile, and trimmed mean. In our numerical experiment, we use the 70% quantile of $\{\lambda_j\}$ as the SPINSURE threshold. This is because the SURE garrote threshold tends to underestimate the true threshold.

We simulate Donoho and Johnstone's four examples with SNR $=$ 5 and apply SURE and SPINSURE procedures to each of the noisy signals. We run 100 simulations. The resulting 100 SURE thresholds and corresponding risks, 100 SPINSURE thresholds and corresponding risks are summarized in Tables 4 and 5. In Table 4 we present the

Table 5. L_2 risks achieved by SURE and SPINSURE thresholds, based on 100 runs of simulations. The noise variance is 1 and the signals are rescaled so that the SNR is 5. *S8* wavelet is used for all examples. For sample size *n*, cycle-spin runs from −n/32+1 to n/32 (totally n/16 times).

Signal			*n = 128*	*n = 256*	*n = 512*	*n = 1024*	*n = 2048*
	Mean	SURE	.726	.564	.410	.290	.194
		SPINSURE	.716	.559	.404	.283	.192
Blocks	Standard	SURE	.035	.017	.019	.014	.005
	Deviation	SPINSURE	.020	.010	.006	.003	.002
	Worst	SURE	.878	.641	.544	.355	.210
	Case	SPINSURE	.825	.598	.421	.294	.201
	Mean	SURE	.702	.608	.456	.315	.205
		SPINSURE	.691	.597	.450	.311	.204
Bumps	Standard	SURE	.034	.021	.013	.010	.004
	Deviation	SPINSURE	.023	.010	.006	.003	.002
	Worst	SURE	.885	.682	.506	.367	.230
	Case	SPINSURE	.803	.639	.476	.325	.212
	Mean	SURE	.284	.187	.125	.075	.047
		SPINSURE	.267	.177	.119	.073	.044
Heavisine	Standard	SURE	.035	.023	.014	.006	.006
	Deviation	SPINSURE	.007	.005	.003	.002	.001
	Worst	SURE	.466	.280	.207	.105	.090
	Case	SPINSURE	.312	.206	.129	.077	.048
	Mean	SURE	.527	.372	.266	.180	.111
		SPINSURE	.514	.358	.262	.176	.108
Doppler	Standard	SURE	.028	.034	.011	.009	.005
	Deviation	SPINSURE	.010	.007	.005	.003	.001
	Worst	SURE	.639	.580	.317	.223	.134
	Case	SPINSURE	.558	.390	.294	.186	.114

average, the smallest, and the biggest threshold for SURE and SPINSURE, respectively. In Table 5, we list the average risk, the standard deviation of the average risk and the worst risk for SURE and SPINSURE, respectively. For sample size $n = 1024$ we use boxplots—see Figure 4—to display the risks.

We have the following observations:

- SPINSURE performs significantly better in terms of L_2 risk.
  This can be seen by conducting a one-sided two-sample hypothesis test:
  $$\text{SURE.risk} = \text{SPINSURE.risk} \quad \text{versus} \quad \text{SURE.risk} > \text{SPINSURE.risk},$$
  and the observed differences in mean risks are all significant (with p-values $\leq$ .005, computed by using large-sample approximation).
- SPINSURE is more stable in the sense of smaller standard deviation.
  This can also be seen from Figure 4: the middle half of the risks are much more compact for SPINSURE.
- SPINSURE improves the worst case over SURE.
- Except for the "heavisine" signal, both SURE and SPINSURE are better than minimax. In Figure 4, we can see that SURE and SPINSURE beat minimax in 594 cases out of 600 simulations.

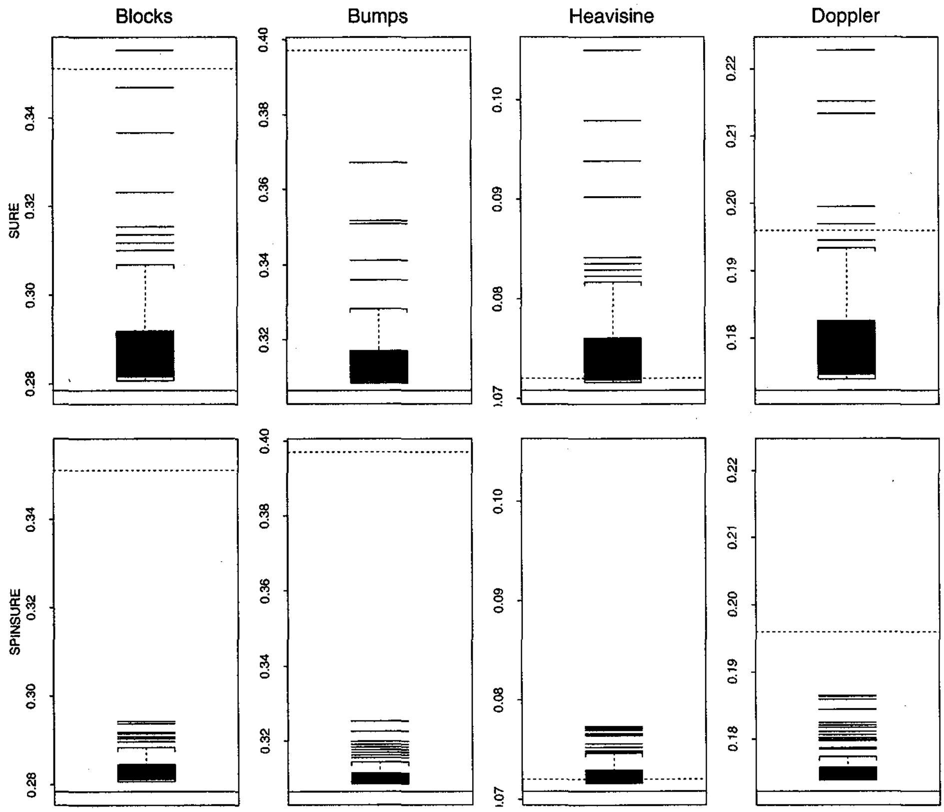

Figure 4. L_2 Risk Comparison Between SURE and SPINSURE Procedures. The shaded box in each plot shows the middle half of the data. The solid horizontal line at the bottom of each plot indicates the crystal-ball risk and the dotted horizontal line indicating the minimax risk. Sample size is 1024 and SNR = 5. The SPINSURE thresholds are more compact than the SURE thresholds.

5. COMPUTATIONAL DETAILS

From Proposition 1, we need to numerically evaluate $A_\lambda(\theta)$ and $B_\lambda(\theta)$ defined by (2.2) and (2.3) respectively. The integrations in (2.2) and (2.3) can be numerically evaluated using standard numerical integration methods listed in Chapter 4 of Press, Teukolsky, Vetterling, and Flannery (1992). Notice that the integrals have infinite upper bounds (referred as "improper" integrals in Press et al. 1992) and two free parameters (λ, θ). To use the methods in Press et al. (1992), one has to choose finite upper bounds and partitions of the finite intervals. For each (λ, θ), the accuracy of the numerical integration depends on the finite upper bound and the partition.

In this article, we propose a different approach. Our method converts an infinite integration problem into an infinite series problem and the latter can be evaluated recursively. To achieve a given precision, one only needs to include enough terms in summations (5.7) and (5.8).

Lemma 1. *The function $A_\lambda(\theta)$ and $B_\lambda(\theta)$ defined in (2.2) and (2.3), respectively, can be written as*

$$A_\lambda(\theta) = 2\theta\phi(\theta)\sum_{n=0}^{\infty}\frac{\theta^{2n}\Gamma_n(\lambda)}{(2n+1)!} \tag{5.1}$$

$$B_\lambda(\theta) = 2\sqrt{2\pi}\phi(\theta)\left\{\frac{\phi(\lambda)}{\lambda} - 1 + \Phi(\lambda)\right\} + 2\phi(\theta)\sum_{n=0}^{\infty}\frac{\theta^{2n+2}\Gamma_n(\lambda)}{(2n+2)!}, \tag{5.2}$$

where

$$\Gamma_n(\lambda) = \int_\lambda^\infty x^{2n}e^{-x^2/2}dx$$

and $\Gamma_n(\lambda)$ can be computed recursively by

$$\Gamma_n(\lambda) = \lambda^{2n-1}e^{-\lambda^2/2} + (2n-1)\Gamma_{n-1}(\lambda), \quad n \geq 1, \tag{5.3}$$

with $\Gamma_0(\lambda) = \sqrt{2\pi}(1-\Phi(\lambda))$.

Proof: Expressions (5.1) and (5.2) can be obtained by expanding the functions $e^{\theta x}$ and $e^{-\theta x}$, and expression (5.3) can be obtained through integration by parts. All the related formulas can be found in Abramowitz and Stegun (1972). □

For numerical stability, let

$$G_n(\lambda) = \frac{\Gamma_n(\lambda)}{(2n-1)!!}, \quad n \geq 0,$$

$$D_n(\theta) = -\frac{\theta^2}{2} + n\log(\theta^2) - \sum_{k=1}^{n}\log(2k), \quad n \geq 0,$$

and

$$C_n(\lambda) = -\frac{\lambda^2}{2} + (2n-1)\log(\lambda) - \sum_{k=1}^{n}\log(2k-1), \quad n \geq 0,$$

where notations $n!! = n(n-2)\dots$ for $n > 0$ and $0!! = (-1)!! = 1$. Then for $n \geq 1$,

$$G_n(\lambda) = G_{n-1}(\lambda) + \exp\{C_n(\lambda)\} \tag{5.4}$$

$$C_n(\lambda) = C_{n-1}(\lambda) + \log\left(\frac{\lambda^2}{2n-1}\right) \tag{5.5}$$

$$D_n(\theta) = D_{n-1}(\theta) + \log\left(\frac{\theta^2}{2n}\right) \tag{5.6}$$

with $C_0(\lambda) = -\lambda^2/2 + \log(\lambda)$, $G_0(\lambda) = \sqrt{2\pi}(1-\Phi(\lambda))$, and $D_0(\theta) = -\theta^2/2 + \log(\theta^2/2)$.

$$A_\lambda(\theta) = \sqrt{2/\pi}\,\theta\sum_{n=0}^{\infty}\frac{G_n(\lambda)\exp\left(D_n(\theta)\right)}{2n+1} \tag{5.7}$$

$$B_\lambda(\theta) = 2\sqrt{2\pi}\phi(\theta)\left\{\frac{\phi(\lambda)}{\lambda} - 1 + \Phi(\lambda)\right\} + \frac{\theta^2}{\sqrt{2\pi}}\sum_{n=0}^{\infty}\frac{G_n(\lambda)\exp\left(D_n(\theta)\right)}{(n+1)(2n+1)}. \tag{5.8}$$

Remark 7. *The infinite series in (5.1) is convergent for any fixed θ; and the infinite series in (5.2) is convergent uniformly in θ and $\lambda > 0$. From standard Γ-function results (see, for example, Abramowitz and Stegun 1972, chap. 6)*

$$\Gamma_n(\lambda) \le \Gamma_n(0) = \sqrt{\frac{\pi}{2}}(2n-1)!!,$$

and

$$n! \ge \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

Hence, in (5.1)

$$2\phi(\theta)\frac{\theta^{2n+2}\Gamma_n(\lambda)}{(2n)!} \le \theta^2 e^{-\theta^2/2}\left(\frac{\theta^2}{2}\right)^n \frac{1}{n!},$$

and in (5.2)

$$\begin{aligned} 2\phi(\theta)\frac{\theta^{2n+2}\Gamma_n(\lambda)}{(2n+2)!} &\le \frac{\sqrt{2\pi}\phi(\theta)}{(2n+1)(n+1)!}\left(\frac{\theta^2}{2}\right)^{n+1} \\ &\le \frac{\sqrt{2\pi}\phi\left(\sqrt{2(n+1)}\right)(n+1)^{n+1}}{(2n+1)(n+1)!} \le \frac{1}{2\sqrt{2\pi}n^{3/2}} \end{aligned}$$

for all θ and $\lambda \ge 0$.

6. CONCLUSIONS AND DISCUSSIONS

In this article, we have proposed a new shrinkage function, the non-negative garrote.

- It is asymptotically equivalent to other methods.
- In small samples, it offers advantages over both the hard shrinkage (generally smaller mean-square-error and less sensitivity to small perturbations in the data) and the soft shrinkage (generally smaller bias and overall mean-square-error), and tighter minimax bounds.
- In small samples, it is comparable to the firm shrinkage, while the latter requires two thresholds.
- SURE and SPINSURE thresholds are available for the garrote shrinkage. They are easy and fast to compute and they are much faster to compute than the firm shrinkage thresholds.
- SPINSURE provides good empirical thresholds that are superior to SURE thresholds and generally better than the minimax thresholds.

The cross-validation threshold selection procedure for the garrote shrinkage can be derived in the same way as for the soft shrinkage in Nason (1995). Pointwise bias and variance study for WaveShrink estimate using garrote shrinkage can be carried out in a straightforward way as in Bruce and Gao (1996b); Marron et al. (1996).

APPENDIX: PROOF OF THEOREM 1

We now show that the non-negative garrote shrinkage estimates have the same near-optimality properties as the hard and the soft shrinkage estimates, as shown by Donoho and Johnstone (1994).

It suffices to show that all the inequalities in section 5 of Donoho and Johnstone (1994) still hold for the non-negative garrote shrinkage.

Lemma 2. *For all θ and $\lambda > 0$,*

$$R_\lambda^G(\theta) \le (\lambda^2+3)\left\{\frac{4\phi(\lambda)}{\lambda} + (\theta^2 \wedge 1)\right\}.$$

Proof: Let $X \sim N(\theta, 1)$. From (4.1),

$$\begin{aligned} R_\lambda^G(\theta) &= E\left\{(X^2-1)I(|X|\le\lambda)\right\} + E\left\{\left(\frac{\lambda^4+2\lambda^2}{X^2}+1\right)I(|X|>\lambda)\right\} \\ &\le (\lambda^2-1)P\left(|X|\le\lambda\right) + (\lambda^2+3)P\left\{|X|>\lambda\right\} \\ &= \lambda^2 + 3P\left\{|X|>\lambda\right\} - P\left(|X|\le\lambda\right) \\ &< \lambda^2+3 \quad \le \quad (\lambda^2+3)\left(1+\frac{4\phi(\lambda)}{\lambda}\right). \end{aligned}$$

On the other hand, since

$$\frac{\lambda^4+2\lambda^2}{x^2} - x^2 \le 2$$

for $|x| > \lambda$, and using the same Taylor expansion trick as Donoho and Johnstone (1994) in proving equation (34) of their article, we can show that

$$P\{|X|>\lambda\} \le \frac{2\phi(\lambda)}{\lambda} + \frac{\theta^2}{4}.$$

Therefore,

$$\begin{aligned} R_\lambda^G(\theta) &= E\left\{X^2-1+\left(\frac{\lambda^4+2\lambda^2}{X^2} - X^2+2\right)I(|X|>\lambda)\right\} \\ &\le \theta^2 + 4P\left\{|X|>\lambda\right\} \\ &\le 2\left(\theta^2+\frac{4\phi(\lambda)}{\lambda}\right) < (\lambda^2+3)\left(\theta^2+\frac{4\phi(\lambda)}{\lambda}\right). \end{aligned}$$

The lemma follows immediately by combining above inequalities. □

Proof: Inequality (3.2) follows from Proposition 2, Lemma 2 and (3.1):

$$\begin{aligned} R\left(\hat{f}^G_\lambda, f\right) &\le \frac{\sigma^2}{2^J} + \frac{\lambda^2+3}{n}\sum_j\sum_k\left(\frac{4\sigma^2\phi(\lambda)}{\lambda} + \theta^2_{jk}\wedge\sigma^2\right) \\ &\le (\lambda^2+3)\left\{R_{\text{oracle}}(\Theta) + \frac{4\sigma^2\phi(\lambda)}{\lambda}\right\}. \end{aligned}$$

□

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